

Ride Share

Overview

RideShare is a web-based carpooling application that connects drivers with passengers traveling in the same route. The platform promotes cost-sharing, reduces traffic congestion, and minimizes carbon emissions. It follows **role-based access** (Admin, Driver, Passenger) with **CRUD operations** implemented in a business-logic-driven manner.

Roles

1. Admin – Can manage drivers and passengers
2. Driver – Can manage rides
3. Passenger – Can book rides

Story Cards

Common Story Cards (Admin, Driver & Passenger)

US01: Role-based login

As an admin/driver/passenger, I should login into the system.

Acceptance Criteria:

1. Admin can login into the system with default credentials.
2. Driver and Passenger can login into the system with the credentials which are saved in database.
3. Upon successful login, redirect the users to respective home pages (admin, driver, passenger) based on the role.
4. Only active user accounts can be logged in to the system.
5. Provide the appropriate error message on the UI, if failed to login.

US20: Logout

As an admin/driver/passenger, I should logout from the system.

Acceptance Criteria:

1. In this story card, user can able to log out from the system.
2. By clicking on logout button, provide the confirmation pop-up whether to logout or not.
3. After confirmation, session should be expired and user must be logged out from the system.
4. The browser cache must be deleted so that the browser's navigation buttons will not work.
5. Provide necessary success / error messages on the UI.

Admin Story Cards

US02: Driver Creation

As an admin, I can create a driver user account in the system.

Acceptance Criteria:

1. Driver user account should be created with default password by adding in the database.
2. The registration_status must be “inactive” by default.
3. Driver ID must be 5-digit random number.
4. No driver should have same email-id and mobile.
5. The system will record a timestamp when a driver is created or modified.
6. Provide necessary success / error messages on the UI.

US03: View registered drivers

As an admin, I can view the drivers records which are registered in the system.

Acceptance Criteria:

1. In this story card, help admin to view details of drivers based on their registration_status.
2. The details of driver to be displayed are: DriverID, DriverName, No.of Rides, Earnings Avg_Rating and registration_status.
3. Add the buttons which helps to sort the data based on Avg_Rating and registration_date.
4. Provide necessary error messages on the UI, if no driver data found in database.

US04: Terminate Drivers

As an admin, I can terminate the drivers with low-performance.

Acceptance Criteria:

1. Provide a button “**Terminate**” in previous story card.
2. If No.of Rides > 10, Earnings > 5000 and Avg_Rating < 3.0, then driver can be terminated by updating the registration_status of driver “**active -> inactive**”.
3. The system will record a timestamp when a driver is created or modified.
4. Provide necessary success / error messages on the UI.

US05: View registered passengers

As an admin, I can view all passenger records which are registered in the system.

Acceptance Criteria:

1. The passenger records should be viewed in descending order based on no.of rides.
2. Filter the passenger details based on their registration_status.
3. Provide necessary success / error messages on the UI.

US06: Ride Approval

As an admin, I can approve the rides which are added by the driver.

Acceptance Criteria:

1. Admin must approve the requests for adding, modifying or deleting the rides by the driver.
2. Upon successful approval, changes must reflect in the database.
3. The system will record a timestamp when a ride is created or modified.
4. Provide necessary success / error messages on the UI.

US07: Admin activity audit trail

As an admin, I need maintain the logs of all CRUD operations (for Driver, Passenger, Rides and Rating data), so that changes are traceable.

Acceptance Criteria:

1. Logs can be stored in 'audit_logs' table.
2. Provide an UI widget for admin to export the 'audit_logs' table as .csv file.
3. Provide necessary success / error messages on the UI.

Driver Story Cards

US08: Register user account (Driver)

As an driver, I can create new user account in the system.

Acceptance Criteria:

1. In this story card, new drivers should be added into the system.
2. Driver ID must be 7-digit random number.
3. License Number must be a 7-digit number which is prefixed with **LI** (eg. **LI6378627**)
4. The system will record a timestamp when a driver is created or modified.
5. Provide necessary success / error messages on the UI.

US09: Create a Ride

As an driver, I can create a ride in the system with admin approval.

Acceptance Criteria:

1. In this story card, add new ride in the database with registration_status as **"inactive"**.
2. After approval of admin, the registration_status of ride can be modified from **"inactive -> active."**
3. The system will record a timestamp when a driver is created or modified.
4. Provide necessary success / error messages on the UI.

US10: View all Rides

As an driver, I can view all the rides which are available in the system.

Acceptance Criteria:

1. In this story card, driver should be able to view the rides based on their registration_status.
2. Provide options to driver to sort the rides in descending order by fare and distance.
3. Provide necessary success / error messages on the UI.

US11: Edit Ride details

As a driver, I can modify the ride details which are available in the system by using race_id.

Acceptance Criteria:

1. In this story card, ride details can be modified in database using race_id.
2. Update the registration_status from “**active** -> **inactive**”.
3. After admin approval, the necessary changes must be reflected in the database by updating registration_status from “**inactive** -> **active**”.
4. The system will record a timestamp when a driver is created or modified.
5. Provide necessary success / error messages on the UI.

US12: Cancel a Ride

As a driver, I can cancel the ride which are available in the system by using race_id.

Acceptance Criteria:

1. In this story card, ride details can be modified in database using race_id.
2. After admin approval, the necessary changes must be reflected in the database by updating registration_status from “**active** -> **inactive**”.
3. The system will record a timestamp when a driver is created or modified.
4. Provide necessary success / error messages on the UI.

US13: Update Profile (Driver)

As a driver, I can update my profile details in the system.

Acceptance Criteria:

1. In this story card, driver profile details can be modified in the database.
2. Apart from Driver ID and License Number, all other details can be modified.
3. Follow necessary input validations while modifying the data.
4. The system will record a timestamp when a driver is created or modified.
5. Provide necessary success / error messages on the UI.

Passenger Story Cards

US14: Register user account (Passenger)

As a passenger, I can create new user account in the system.

Acceptance Criteria:

1. In this story card, new passenger records should be added into the database.
2. Passenger ID must be 7-digit random number.
3. No two passengers should have same email id and mobile.
4. Add necessary input validations for Name, email, password, mobile and city.
5. The system will record a timestamp when a driver is created or modified.
6. Provide necessary success / error messages on the UI.

US15: Browse rides

As an passenger, I can browse the active rides which are available in the system.

Acceptance Criteria:

1. In this story card, provide the option for passenger to view all registered active rides from the database.
2. Display those rides in grid format on the UI Page.
3. Display the rides on the UI page with necessary Driver and Ride details.
4. Provide necessary success / error messages on the UI.

US16: Book a ride

As a passenger, I can book an active ride which are available in the system.

Acceptance Criteria:

1. In this story card, provide the option for passenger to book the ride.
2. Once ride is book, those ride details should be displayed in **Bookings History**.
3. Provide the option for passenger to view the latest booking by sorting with booking date.
4. The system will record a timestamp when a driver is created or modified.
5. Provide necessary success / error messages on the UI.

US17: Rating Driver

As a passenger, I can rate the driver after ride completion in scale 1.0 to 5.0.

Acceptance Criteria:

1. In the story card, the ride status should be modified in the database.
2. The possible ride status are **Not Started, In Progress, Completed**.
3. Passenger can rate the driver for that ride with ride_status as **completed**.
4. The system will record a timestamp when a driver is created or modified.
5. Provide necessary success / error messages on the UI.

US18: Update Profile (Passenger)

As a passenger, I can update my profile details in the system.

Acceptance Criteria:

1. In this story card, passenger profile details can be modified in the database.
2. Apart from Passenger ID, all other details can be modified.
3. Follow necessary input validations while modifying the data.
4. The system will record a timestamp when a driver is created or modified.
5. Provide necessary success / error messages on the UI.

US19: Deactivate account

As a passenger, I can deactivate my user account in the system.

Acceptance Criteria:

1. In this story card, the registration_status of the passenger can be modified from “**active** -> **inactive**”.
2. Once deactivated, passenger can't able to login into the system.
3. The system will record a timestamp when a driver is created or modified.
4. Provide necessary success / error messages on the UI.

Database Schema

Passengers	
Field	Type
PassengerID	INT
PassengerName	VARCHAR(30)
Email	VARCHAR(30)
password	VARCHAR(30)
mobile	BIGINT
city	VARCHAR(30)
Registration_status	VARCHAR(10)
Created_on	TIMESTAMP
Modified_on	TIMESTAMP

Drivers	
Field	Type
DriverID	INT
DriverName	VARCHAR(30)
Email	VARCHAR(30)
Password	VARCHAR(30)
Mobile	BIGINT
LicenseNumber	INT
VehicleMake	VARCHAR(30)
VehicleModel	VARCHAR(30)
Available_seats	INT
Registration_Status	VARCHAR(10)
Created_on	TIMESTAMP

Modified_on	TIMESTAMP
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Rides	
Field	Type
Ride_id	INT
Source	VARCHAR(30)
Destination	VARCHAR(30)
Depature_time	TIMESTAMP
Distance (in Kms)	INT
Fare (per seat)	DECIMAL(7, 2)
Ride_status	VARCHAR(10)
Registration_Status	
Created_on	TIMESTAMP
Modified_on	TIMESTAMP

Bookings	
Field	Type
Booking_id	INT
Seats_booked	INT
Booking_status	VARCHAR(10)
Created_on	TIMESTAMP
Modified_on	TIMESTAMP

Ratings	
Field	Type
Rating_id	INT
Rating_Value	DECMIAL(2,1)
Created_on	TIMESTAMP
Modified_on	TIMESTAMP

Audit Logs	
Field	Type
Log_id	INT
Action (INSERT, UPDATE , DELETE)	VARCHAR(10)
Entity_type (Passenger, Driver, Ride, Bookings, Rating)	VARCHAR(15)
Description	VARCHAR(100)
Created_on	TIMESTAMP